PRAKTIKUM BASIS DATA

MODUL 8

08_Scripting_SQL_Create

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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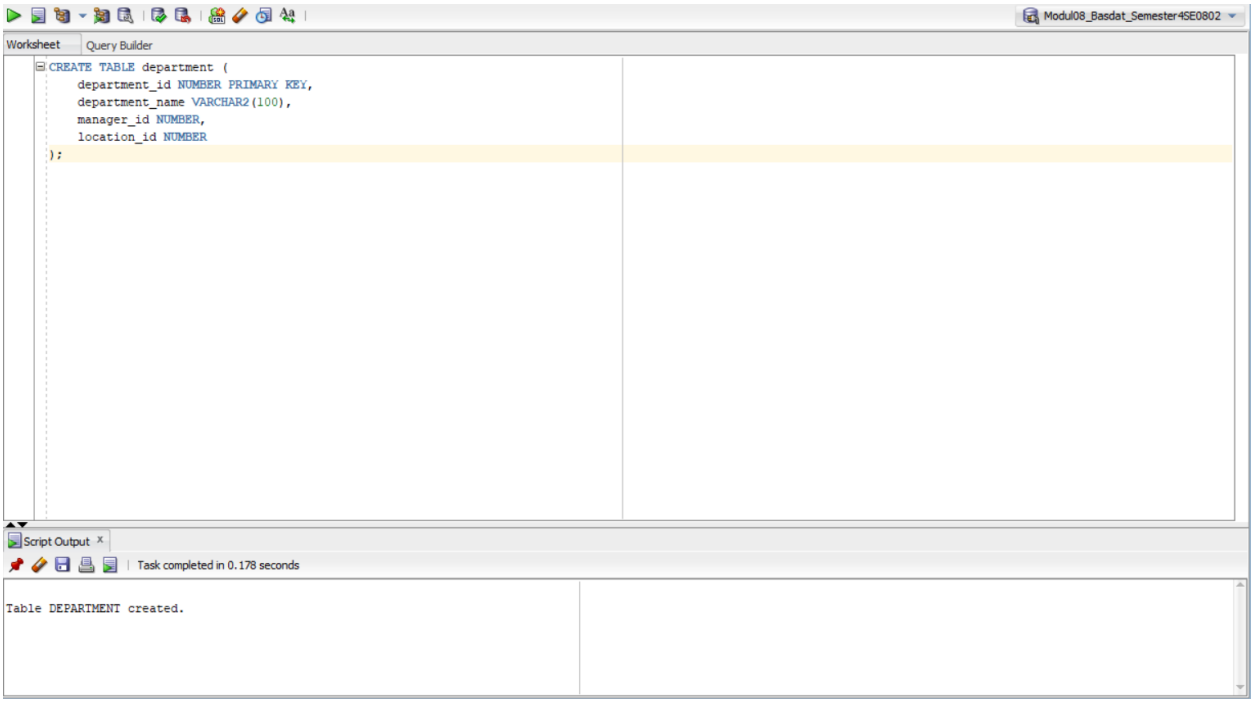
UNIVERSITAS TELKOM

2026

Jurnal

1. Buatlah desain database untuk suatu perusahaan:

   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.



   b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>

```
CREATE TABLE department (
    department_id NUMBER PRIMARY KEY,
    department_name VARCHAR2(100),
    manager_id NUMBER,
    location_id NUMBER
);

CREATE TABLE employee (
    employee_id NUMBER PRIMARY KEY,
    last_name VARCHAR2(100) NOT NULL,
    email VARCHAR2(100),
    salary NUMBER,
    commission_pct NUMBER,
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_department
        FOREIGN KEY (department_id)
        REFERENCES department(department_id)
);
```

```
Table DEPARTMENT created.

Table EMPLOYEE created.
```

c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).

```
    department_name VARCHAR2(100),
    manager_id NUMBER,
    location_id NUMBER
);

CREATE TABLE employee (
    employee_id NUMBER PRIMARY KEY,
    last_name VARCHAR2(100) NOT NULL,
    email VARCHAR2(100),
    salary NUMBER,
    commission_pct NUMBER,
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_department
        FOREIGN KEY (department_id)
        REFERENCES department(department_id)
);

CREATE TABLE supplier (
    supplier_id NUMBER PRIMARY KEY,
    supplier_name VARCHAR2(100),
    department_id NUMBER,
    CONSTRAINT fk_supplier_department
        FOREIGN KEY (department_id)
        REFERENCES department(department_id)
);
```

```
Table EMPLOYEE created.

Table SUPPLIER created.
```

2. Analisis table-table dengan compute statistics

```
    email VARCHAR2(100),
    salary NUMBER,
    commission_pct NUMBER,
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_department
        FOREIGN KEY (department_id)
        REFERENCES department(department_id)
);

CREATE TABLE supplier (
    supplier_id NUMBER PRIMARY KEY,
    supplier_name VARCHAR2(100),
    department_id NUMBER,
    CONSTRAINT fk_supplier_department
        FOREIGN KEY (department_id)
        REFERENCES department(department_id)
);

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
```

Script Output — Task completed in 0.077 seconds

```
        or drop the existing object so it can be reused.

Table DEPARTMENT analyzed.


Table EMPLOYEE analyzed.


Table SUPPLIER analyzed.
```

3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).

```
    supplier_id NUMBER PRIMARY KEY,
    supplier_name VARCHAR2(100),
    department_id NUMBER,
    CONSTRAINT fk_supplier_department
        FOREIGN KEY (department_id)
        REFERENCES department(department_id)
);

INSERT ALL
    INTO department VALUES (10, 'HR', 100, 1700)
    INTO department VALUES (20, 'Finance', 101, 1701)
    INTO department VALUES (30, 'IT', 102, 1702)
    INTO department VALUES (40, 'Marketing', 103, 1703)
    INTO department VALUES (50, 'Sales', 104, 1704)
    INTO department VALUES (60, 'Logistics', 105, 1705)
    INTO department VALUES (70, 'Support', 106, 1706)
    INTO department VALUES (80, 'Operations', 107, 1707)
    INTO department VALUES (90, 'Legal', 108, 1708)
    INTO department VALUES (100, 'R&D', 109, 1709)
SELECT * FROM DUAL;

INSERT ALL
```

Script Output — Task completed in 0.026 seconds

```
https://docs.oracle.com/error-help/db/ora-00001/

More Details :
https://docs.oracle.com/error-help/db/ora-00001/

Table DEPARTMENT truncated.


10 rows inserted.
```

```
    INTO department VALUES (70, 'Support', 106, 1706)
    INTO department VALUES (80, 'Operations', 107, 1707)
    INTO department VALUES (90, 'Legal', 108, 1708)
    INTO department VALUES (100, 'R&D', 109, 1709)
SELECT * FROM DUAL;

INSERT ALL
    INTO employee VALUES (1, 'Andi', 'andi@mail.com', 5000, 0.1, SYSDATE, 80)
    INTO employee VALUES (2, 'Budi', 'budi@mail.com', 6000, 0.2, SYSDATE, 80)
    INTO employee VALUES (3, 'Citra', 'citra@mail.com', 7000, NULL, SYSDATE, 10)
    INTO employee VALUES (4, 'Dewi', 'dewi@mail.com', 5500, 0.15, SYSDATE, 20)
    INTO employee VALUES (5, 'Eka', 'eka@mail.com', 4500, NULL, SYSDATE, 30)
    INTO employee VALUES (6, 'Fajar', 'fajar@mail.com', 8000, 0.3, SYSDATE, 40)
    INTO employee VALUES (7, 'Gina', 'gina@mail.com', 5200, NULL, SYSDATE, 50)
    INTO employee VALUES (8, 'Hadi', 'hadi@mail.com', 6200, 0.2, SYSDATE, 60)
    INTO employee VALUES (9, 'Intan', 'intan@mail.com', 7100, NULL, SYSDATE, 70)
    INTO employee VALUES (10, 'Joko', 'joko@mail.com', 9000, 0.25, SYSDATE, 80)
SELECT * FROM DUAL;

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
```

Script Output — Task completed in 0.023 seconds

```
https://docs.oracle.com/error-help/db/ora-00001/

Table DEPARTMENT truncated.

10 rows inserted.

10 rows inserted.
```

```
    INTO employee VALUES (4, 'Dewi', 'dewi@mail.com', 5500, 0.15, SYSDATE, 20)
    INTO employee VALUES (5, 'Eka', 'eka@mail.com', 4500, NULL, SYSDATE, 30)
    INTO employee VALUES (6, 'Fajar', 'fajar@mail.com', 8000, 0.3, SYSDATE, 40)
    INTO employee VALUES (7, 'Gina', 'gina@mail.com', 5200, NULL, SYSDATE, 50)
    INTO employee VALUES (8, 'Hadi', 'hadi@mail.com', 6200, 0.2, SYSDATE, 60)
    INTO employee VALUES (9, 'Intan', 'intan@mail.com', 7100, NULL, SYSDATE, 70)
    INTO employee VALUES (10, 'Joko', 'joko@mail.com', 9000, 0.25, SYSDATE, 80)
SELECT * FROM DUAL;

INSERT ALL
    INTO supplier VALUES (1, 'Supplier A', 10)
    INTO supplier VALUES (2, 'Supplier B', 20)
    INTO supplier VALUES (3, 'Supplier C', 30)
    INTO supplier VALUES (4, 'Supplier D', 40)
    INTO supplier VALUES (5, 'Supplier E', 50)
    INTO supplier VALUES (6, 'Supplier F', 60)
    INTO supplier VALUES (7, 'Supplier G', 70)
    INTO supplier VALUES (8, 'Supplier H', 80)
    INTO supplier VALUES (9, 'Supplier I', 90)
    INTO supplier VALUES (10, 'Supplier J', 100)
SELECT * FROM DUAL;
```

Script Output — Task completed in 0.03 seconds

```
10 rows inserted.

10 rows inserted.

10 rows inserted.
```

4. Analisis table-table dengan compute statistics

```
    INTO employee VALUES (7, 'Gina', 'gina@mail.com', 5200, NULL, SYSDATE, 50)
    INTO employee VALUES (8, 'Hadi', 'hadi@mail.com', 6200, 0.2, SYSDATE, 60)
    INTO employee VALUES (9, 'Intan', 'intan@mail.com', 7100, NULL, SYSDATE, 70)
    INTO employee VALUES (10, 'Joko', 'joko@mail.com', 9000, 0.25, SYSDATE, 80)
SELECT * FROM DUAL;

INSERT ALL
    INTO supplier VALUES (1, 'Supplier A', 10)
    INTO supplier VALUES (2, 'Supplier B', 20)
    INTO supplier VALUES (3, 'Supplier C', 30)
    INTO supplier VALUES (4, 'Supplier D', 40)
    INTO supplier VALUES (5, 'Supplier E', 50)
    INTO supplier VALUES (6, 'Supplier F', 60)
    INTO supplier VALUES (7, 'Supplier G', 70)
    INTO supplier VALUES (8, 'Supplier H', 80)
    INTO supplier VALUES (9, 'Supplier I', 90)
    INTO supplier VALUES (10, 'Supplier J', 100)
SELECT * FROM DUAL;

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
```

```
Table DEPARTMENT analyzed.

Table EMPLOYEE analyzed.

Table SUPPLIER analyzed.
```

5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.

```
    INTO supplier VALUES (3, 'Supplier C', 30)
    INTO supplier VALUES (4, 'Supplier D', 40)
    INTO supplier VALUES (5, 'Supplier E', 50)
    INTO supplier VALUES (6, 'Supplier F', 60)
    INTO supplier VALUES (7, 'Supplier G', 70)
    INTO supplier VALUES (8, 'Supplier H', 80)
    INTO supplier VALUES (9, 'Supplier I', 90)
    INTO supplier VALUES (10, 'Supplier J', 100)
SELECT * FROM DUAL;

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;

CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;
```

```
Table EMPLOYEE analyzed.

Table SUPPLIER analyzed.

View EMPVU80 created.
```

6. Kemudian tampilkan struktur dari view empvu80.

```
INTO supplier VALUES (5, 'Supplier E', 50)
INTO supplier VALUES (6, 'Supplier F', 60)
INTO supplier VALUES (7, 'Supplier G', 70)
INTO supplier VALUES (8, 'Supplier H', 80)
INTO supplier VALUES (9, 'Supplier I', 90)
INTO supplier VALUES (10, 'Supplier J', 100)
SELECT * FROM DUAL;

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;

CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;

DESC empvu80;
```

Script Output — Task completed in 0.439 seconds

```
View EMPVU80 created.

Name          Null?    Type
------------- -------- -------------
ID_NUMBER     NOT NULL NUMBER
NAME          NOT NULL VARCHAR2(100)
SALARY                 NUMBER
DEPARTMENT_ID          NUMBER
```

7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE

```
INTO supplier VALUES (8, 'Supplier H', 80)
INTO supplier VALUES (9, 'Supplier I', 90)
INTO supplier VALUES (10, 'Supplier J', 100)
SELECT * FROM DUAL;

ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;

CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;

DESC empvu80;

CREATE INDEX idx_emp_dept
ON employee(department_id);
```

Script Output — Task completed in 0.027 seconds

```
Name          Null?    Type
------------- -------- -------------
ID_NUMBER     NOT NULL NUMBER
NAME          NOT NULL VARCHAR2(100)
SALARY                 NUMBER
DEPARTMENT_ID          NUMBER

Index IDX_EMP_DEPT created.
```